

12.775

EE25BTECH11033 - Kavın

Question:

Choose solution set S corresponding to the systems of two equations

$$x - 2y + z = 0$$

$$x - z = 0$$

$$1) S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\}$$

$$3) S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} \middle| \alpha, \beta \in \mathbb{R} \right\}$$

$$2) S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha, \beta \in \mathbb{R} \right\}$$

$$4) S = \left\{ \alpha \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \middle| \alpha \in \mathbb{R} \right\}$$

Solution:

We are given a homogeneous system of two linear equations with three variables (x , y , z). We need to find all vectors $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ that satisfy both equations simultaneously.

We can solve this system using substitution.

- **From the second equation:**

$$x - z = 0 \implies x = z \quad (1)$$

- **Substitute $x = z$ into the first equation:**

$$(z) - 2y + z = 0 \quad (2)$$

$$2z - 2y = 0 \quad (3)$$

$$2z = 2y \quad (4)$$

$$z = y \quad (5)$$

- **Combine the results:** We have found that $x = z$ and $y = z$. Therefore, all three variables must be equal:

$$x = y = z \quad (6)$$

Since we have three variables (x, y, z) but only two independent equations, we have one free variable. We can express the solution set in terms of a single parameter, let's call it $\alpha \in \mathbb{R}$.

- Let the free variable be $z = \alpha$.

- Since $y = z$, then $y = \alpha$.
- Since $x = z$, then $x = \alpha$.

The solution vectors are of the form:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \alpha \\ \alpha \\ \alpha \end{pmatrix} \quad (7)$$

We can factor out the parameter α :

$$\alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (8)$$

The solution set S is the set of all possible vectors of this form for any real number α .

Therefore, the solution set is:

$$S = \left\{ \alpha \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \mid \alpha \in \mathbb{R} \right\} \quad (9)$$

